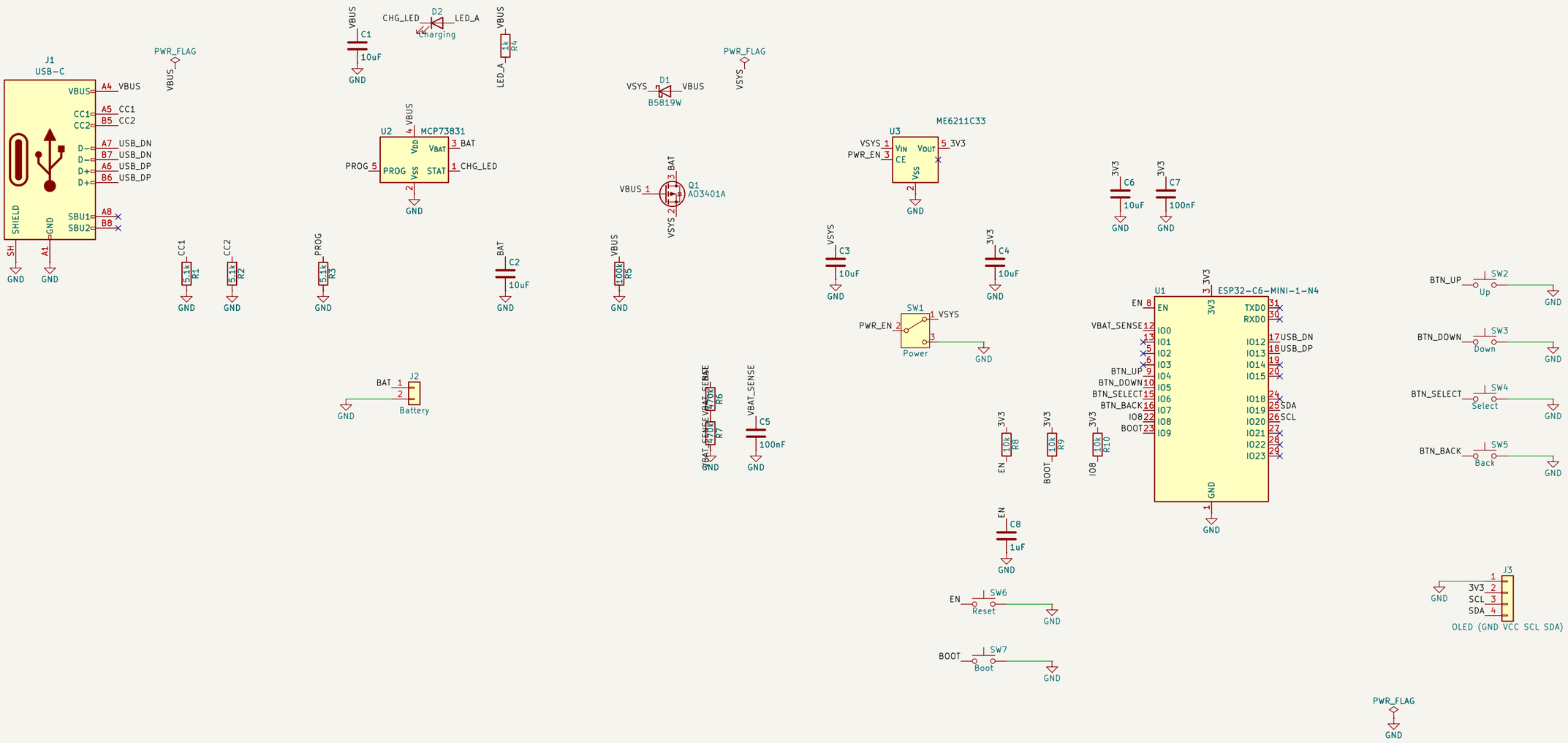
UtilWatch board

Factory-assembled (JLCPCB parts, LCSC numbers in each part's LCSC field). You solder the OLED and plug in the battery.
Same GPIOs as the breadboard build: buttons IO4–7 to GND, OLED SDA IO19 / SCL IO20. Flash over USB-C (IO12/IO13).



Power:
- USB plugged in: D1 feeds VSYS and Q1 is off, so the charger only charges the battery.
- Unplugged: Q1 connects the battery to VSYS.
- SW1 switches the 3.3 V regulator. Charging works with SW1 off; the watch (and USB flashing) needs SW1 on.
- Use a LiPo with its own protection board (most hobby cells have one). Check the JST plug's polarity: pin 1 is +.

ESP32-C6-MINI-1, factory assembled

Sheet: /
File: utilwatch.kicad_sch

Title: UtilWatch board

Size: A3 | Date: | **Rev: 0.1**
KiCad E.D.A. 10.0.8 | Id: 1/1